Unit 12, Lesson 7: Evaluating Expressions with Radicals

Benchmark

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

Learning Target

- ☐ I can solve problems involving the order of operations with rational numbers, including exponents and radicals.

12.7.1 Warm-Up: Square Root Expression

1. Place a digit in each blue box to create an expression with a value as close to 0 as possible. Use only the digits 0 through 9, at most one time each.

$$\sqrt{\square\square} - \sqrt{\square\square} + \sqrt{\square}$$

   a. Describe your strategy in determining where to put the numbers.

   b. What is the value of your expression?

   c. Create another expression that is as close or closer to 0 than your first.

12.7.2 Refresher: Order of Operations

Recall that the order of operations is an agreed-upon set of rules for how to evaluate an expression.

1. **Work with your partner to number each operation so that the operations are in the correct order.**

◯	Perform multiplication and division in order from left to right. (→)
◯	Evaluate expressions with exponents.
◯	Perform operations inside grouping symbols.
◯	Perform addition and subtraction in order from left to right. (→)

2. **Based on what you have learned about radicals, discuss with your partner where you think evaluating radicals falls within the order of operations. Then, write a summary of your discussion.**

12.7.3 Guided Instruction: Evaluating Expressions With Radicals

Evaluating exponents and radicals falls within the same step in the order of operations, where each is evaluated from left to right, after operations within grouping symbols are performed.

1. **Simon evaluated the expression $\frac{\left(|3-\sqrt{25}|^3\right)}{16}$, as shown.**

Original Expression	$\frac{\left(|3-\sqrt{25}|^3\right)}{16}$
Step 1	$\frac{|3-5|^3}{16}$
Step 2	$\frac{|-2|^3}{16}$
Step 3	$\frac{2^3}{16}$
Step 4	$\frac{8}{16}=\frac{1}{2}$

a. **With your partner, discuss the steps Simon took to find the value of the expression.**

b. **Which grouping symbols are used in the expression?**

c. Use the words or phrases in the bank to complete the statements.

absolute value	equivalent	exponent
grouping symbols	radicals	subtraction

Simon started by evaluating $3-\sqrt{25}$ because the absolute value bars are ____________________. Within the bars, Simon first evaluated $\sqrt{25}$ because evaluating ________________ comes before ________________. He was able to evaluate the ________________ only after finding the value of the grouped ____________________ expression. Finally, he found an ________________ fraction for the expression that remained.

2. Find the value of the expression. Show your work.

$$\left(\frac{1}{2}\right)^{-2}+\sqrt{4^2+9}$$

3. **Simon, Tonya, and Cassandra evaluated the following expression using substitution and the order of operations. Their work is shown in the table.**

$$\sqrt{56.5 - b} - 4.1^2, \text{ where } b = -7.5$$

Simon	Tonya	Cassandra
$\sqrt{56.5 - b} - 4.1^2$	$\sqrt{56.5 - b} - 4.1^2$	$\sqrt{56.5 - b} - 4.1^2$
$\sqrt{56.5 - 7.5} - 4.1^2$	$\sqrt{56.5 + 7.5} - 4.1^2$	$\sqrt{56.5 + 7.5} - 4.1^2$
$\sqrt{49} - 4.1^2$	$\sqrt{64} - 4.1^2$	$\sqrt{64} - 4.1^2$
$7 - 4.1^2$	$8 - 4.1^2$	$8 - 4.1^2$
$7 - 16.81$	$8 + 16.81$	$8 - 16.81$
-9.81	24.81	-8.81

a. **Discuss each student's work with your partner.**

b. **Circle the steps where you notice differences in the students' work.**

c. **Which student correctly evaluated the expression?**

d. Complete the table to describe the error(s) each of the other two students made.

Student's Name		
Error(s) Made When Evaluating		

12.7.4 Guided Practice: Evaluating Expressions With Radicals

1. Find the value of the expression. Show your work.

$$\frac{18 - 2 \cdot \sqrt{121}}{2^2}$$

2. Find the value of the expression $\frac{6.2+\sqrt[3]{x}}{12}$, where $x = -27$.

12.7.5 Your Turn!

Find the value of each expression. Show your work.

1. $\frac{5 \cdot 3^4}{-\sqrt{225}}$

2. $\sqrt[3]{100 - 6^2}$

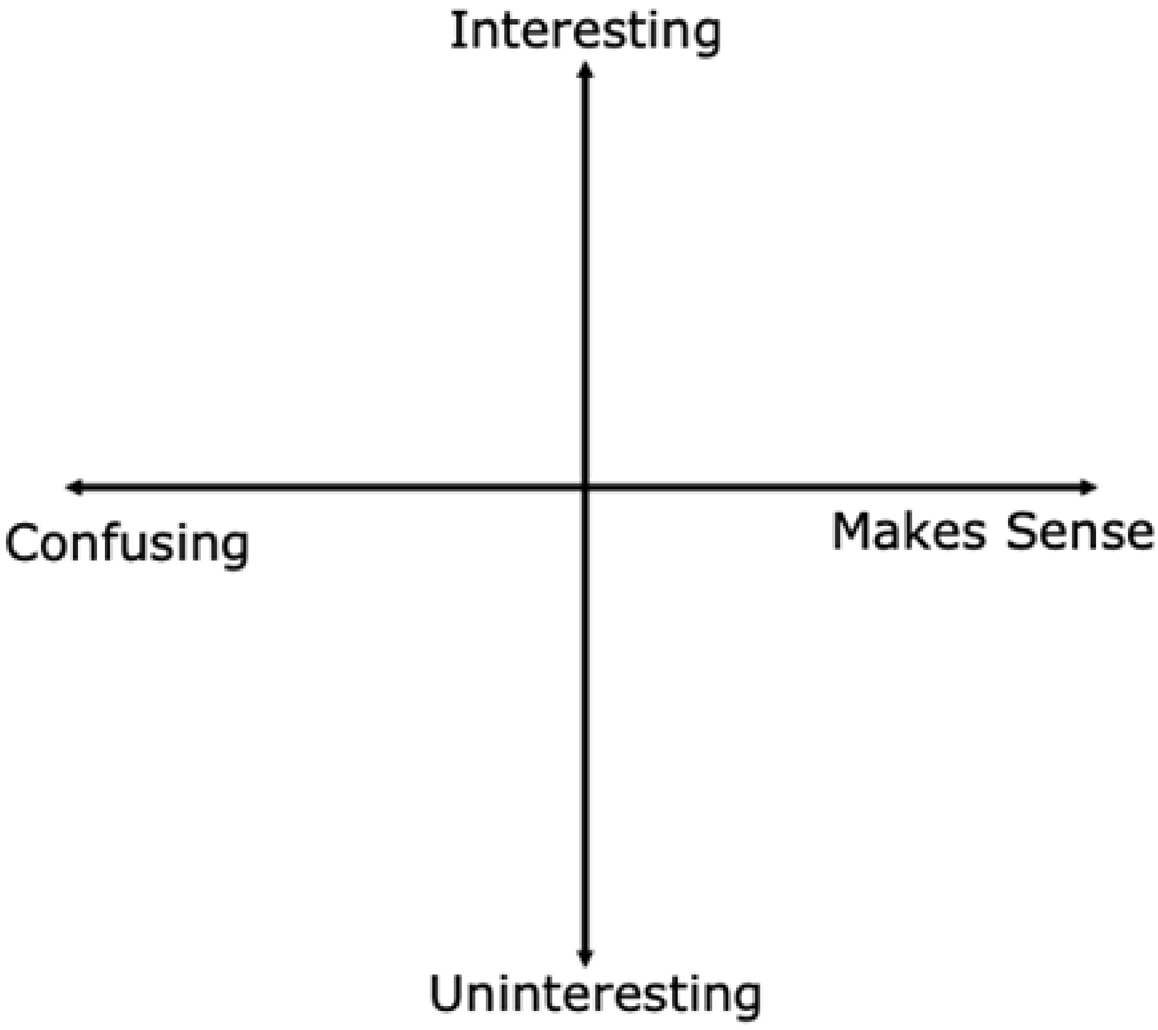
3. $\left| 4^{-1} + \sqrt[3]{\frac{1}{8}} \right|$

4. $\frac{\sqrt{81} - w^2}{9} + 25$ where $w = -4$

12.7.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how you feel about today's learning target.

- ☐ **I can solve problems involving the order of operations with rational numbers, including exponents and radicals.**



2. **Write one question you still have about this learning target after today's lesson.**